

Quizora: AI-Powered Examination & Learning Platform

Project Report

1. Project Overview

Quizora is an intelligent, full-stack web application designed to facilitate both adaptive learning and proctored examinations. By leveraging Artificial Intelligence and modern web technologies, Quizora provides a dynamic educational experience tailored to individual student needs while ensuring the integrity of formal assessments.

2. Technology Stack

Backend: Python with Flask framework

Database: SQLite (quizora.db)

Frontend: HTML5, vanilla JavaScript, CSS (with modern UI aesthetics including glassmorphism and dynamic animations)

AI & Machine Learning:

Generative AI for dynamic question generation, personalized feedback, hints, and explanations (ai_logic.py).

PDF parsing (pypdf) for automatic quiz generation from uploaded study materials.

Browser-based AI models (face-api.js) for webcam-based proctoring and face detection.

3. Key Features

3.1 Adaptive Learning Mode

Dynamic Question Generation: AI generates questions on-the-fly based on the selected category and the student's current difficulty level.

Adaptive Difficulty: The system automatically adjusts the difficulty of the next question based on whether the student answered the previous question correctly.

AI Assistance: Students can request AI-generated hints and detailed explanations for questions to enhance their understanding.

PDF to Quiz: Users can upload PDF documents, and the system uses AI to extract text and automatically generate a relevant quiz.

3.2 Formal Examination System & Proctoring

Exam Management: Admins can create exams, set time limits, total marks, and passing criteria.

Anti-Cheating Mechanisms:

- **Tab Switching Detection:** Logs and warns users if they navigate away from the exam tab.
- **Fullscreen Enforcement:** Requires the exam to be taken in fullscreen mode. Exiting triggers a warning.
- **Webcam Monitoring:** Uses the device's webcam to monitor the student. An event listener detects if the webcam is disconnected during the exam.
- **Input Restriction:** Disables right-clicking and text copying to prevent plagiarism.
- **Three-Strike Rule:** Automatically terminates the exam and scores a 0 if the maximum number of cheating warnings is reached.

3.3 Gamification & Engagement

- **Points & Streaks:** Students earn points for completing quizzes. Consecutive daily logins build up a "current streak" and "longest streak".
- **Badges:** Achievement badges are awarded for milestones (e.g., 'First Steps', 'Quick Learner', 'Steady Scholar', 'Master Mind').
- **Leaderboard:** Displays the top 10 students based on their earned points, encouraging healthy competition.

3.4 Role-Based Access Control

- **Students:** Can access the learning dashboard, take quizzes/exams, view their history, and see their stats/badges.
- **Administrators:** Have access to an admin dashboard to manage exams, add/delete questions, view aggregate student records, and clear history.

4. System Architecture

The application follows a standard Client-Server architecture:

- **Frontend (Client):** Renders the user interface using Jinja2 templates (frontend/templates). Static assets (CSS, JS, downloaded ML models) are served from frontend/static.

- **Backend (Server):** The Flask application (backend/app.py) handles routing, session management, database interactions, and API endpoints.
- **AI Engine:** The backend communicates with external LLM APIs (via ai_logic.py) to generate content.
- **Database:** A lightweight SQLite database stores users, questions, exams, quiz results, and gamification data.

5. Security & Privacy

- **Password Hashing:** User passwords are securely hashed using werkzeug.security before being stored in the database.
- **Session Management:** Secure Flask sessions determine user authentication and authorization (Admin vs. Student).
- **Local Processing for Proctoring:** Webcam streams are processed locally in the browser to detect tampering, ensuring privacy by not continuously transmitting video feeds to the server (unless specifically logged).

6. Conclusion

Quizora represents a modern approach to digital education, seamlessly blending rigorous assessment protocols with an engaging, AI-driven learning environment. The inclusion of strict proctoring tools makes it suitable for formal academic use, while the gamified learning mode ensures students remain motivated and challenged.